Values of $L(x)$ are estimated using its relationship with $l(x)$ and $d(x)$ as follows:

$$L(83) = \frac{l(83) + l(84)}{2}$$

$$= \frac{3560 + 2990}{2}$$

$$= 3{,}275$$

$$L(84) = l(84) - \frac{1}{2} d(84)$$

$$2990 - \frac{508}{2}$$

$L(84) = 2736.$

The value of $T(83)$ can be estimated from the given value of $T(84)$ and the estimated value of $L(83)$ from the relationship

$$T(84) = T(83) - L(83)$$

$$T(83) = L(83) + T(84)$$

as

$$T(83) = 3{,}275 + 11975 = 15{,}250$$

The life expectancy of the ***cohort*** at the age x = 83 and 84 years is estimated using the relationship $e^0(x) = \frac{T(x)}{l(x)}$

as

$$e^0(83) = \frac{15250}{3560} = 4.28$$

$$e^0(84) = \frac{11975}{2990} = 4.01$$

Now, the completed Life Table is as follows:

x	$l(x)$	$d(x)$	$p(x)$	$q(x)$	$L(x)$	$T(x)$	$e^0(x)$
83	3560	570	0.84	0.16	3275	15250	4.28
84	2990	508	0.83	0.17	2736	11975	4.01

Example 8.8

A part of the Life Table of a population is given hereunder with incomplete information. Find those information from the given data and complete the Life Table.

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